Exercises

Exercise 1.1 · Algebraic form. Write in algebraic form:

1. $(3-2i)+(-1+5i)$;
2. $(3-2i)(-1+5i)$;
3. $(1-2i)^2$;
4. i^{2026}.

Exercise 1.2 · Inverses and quotients. Write in algebraic form: $\dfrac{1}{1+i}$; $\dfrac{2+i}{1-2i}$; $\dfrac{3+i}{2-i}$.

Exercise 1.3 · Equations. Solve in $\mathbb{C}$:

1. $(1+i)z = 3-i$;
2. $2z - 3\overline{z} = 1+10i$;
3. $z + 2\overline{z} = 6-i$.

Exercise 1.4 · Real part and imaginary part of an expression. For $z = x+iy$ (x, y real), let $Z = z^2 - 2\overline{z}$.

1. Express $\mathrm{Re}(Z)$ and $\mathrm{Im}(Z)$ in terms of x and y.
2. Determine the complex numbers z for which Z is real.

Exercise 1.5 · Binomial formula.

1. Using the binomial formula, write $(1+i)^5$ in algebraic form.
2. Deduce $(1+i)^{10}$.
3. Recover these two results from $(1+i)^2$.

Exercise 1.6 · A purely imaginary quotient. Let z be a complex number such that $z\overline{z} = 1$ and $z \neq 1$. Let $w = \dfrac{1+z}{1-z}$.

1. Explain why $\overline{z} = \dfrac{1}{z}$.
2. Compute $\overline{w}$ and prove that w is purely imaginary.
3. Check the result for $z = i$.

Exercise 1.7 · Sum of the powers of i.

1. Prove that, for every complex number $q \neq 1$ and every natural number n, $1+q+\cdots+q^n = \dfrac{1-q^{n+1}}{1-q}$.
2. Deduce $S = 1+i+i^2+\cdots+i^{2026}$.

3. Recover this result by grouping the terms in blocks of four.